

FINITE-SAMPLE SAFE MODEL RANKING UNDER SUBGROUP-MIX TURNOVER

Anonymous authors

Paper under double-blind review

ABSTRACT

At deployment the class/subgroup mixture of a fixed pool of candidate models can shift (label-prior / subgroup-mix turnover) while each subgroup’s class-conditional risk stays unchanged, so a model optimal for the collected mixture may cease to be optimal—and a worse model can be selected. We formalize *safe ranking* as a finite-sample certification problem: given a calibration split, decide for every future mixture weight w whether to *commit* a model with a certificate that its regret (loss minus the oracle-best model’s loss at w) is at most an operator-set tolerance τ , and otherwise *abstain* honestly rather than hard-select a possibly-overturning pick. We prove a linear-band structure (P1) giving joint coverage $\geq 1 - \delta$ of every mixture risk by weighted Clopper–Pearson bands, and a tighter *paired-difference* (MCB-style) certificate (P3/Thm. 1) that bounds $\max_j (R_{i^*} - R_j)$ by pairing models on the *same* calibration points, cancelling shared error. On four public carriers and four classifiers, certified commits always have true regret $\leq \tau$ (coverage = 1.0), and naively *minimizing the upper band* is shown to be a vacuous or *anti-selective* selector under heteroskedastic group widths—so we use *point-estimate selection + certificate gating* (M2/M2.5), abstaining exactly on turnover-skew mixtures where hard-selection pays orders-of-magnitude higher regret. The paired certificate exposes an explicit *certificate-price frontier*: exact finite-sample bounds commit 0.269–0.260 (matched 5-seed means), a diagnostic asymptotic CI raises this to 0.503, at no loss of soundness. Under a *label budget* (Sec. 6) soundness persists (0 violations over 4200 cells); uniform spread out-commits variance-aware rules on MNIST but is beaten by them on Fashion at low budget, and a time-uniform adaptive certificate charges an adaptivity width no carrier repays. A fixed-budget minimax analysis (Sec. 7) pins down *when* uniform is the optimal allocation (a verifiable symmetry condition) and *when* a convex-minimax rule strictly dominates, with a conditional gate whose driver is whether the deployed mixture set spans the subgroups.

1 INTRODUCTION

Model selection is usually done once, on the mixture where data was collected. But at deployment the label/subgroup mix can move—a model trained on general web traffic now serves a skewed query stream; a classifier tuned on balanced data is deployed onto a single class-dominant population. Under *label-prior shift* (*subgroup-mix turnover*) the class-conditional distribution $P(X \mid Y=g)$ is fixed but the mixture weight w over subgroups is not. A model i ’s risk is then linear in w ,

$$R_i(w) = \sum_g w_g r_{i,g}, \quad r_{i,g} = \mathbb{E}[\ell(\text{pred}_i, Y) \mid Y=g], \quad (1)$$

so the identity of the best model, $\arg \min_i R_i(w)$, can change as w moves across the simplex. Standard selection heuristics live at one of two extremes: *single-point* selection (pick by the collected-mixture estimate) ignores w entirely; *worst-group* / *DRO* / *balanced-accuracy* selection ignores where w actually is. None answers the deployment question: *for a given future mixture w , can I both pick a model and certify its regret in finite sample—and when must I abstain rather than risk a pick that overturns $w \mapsto w'$?*

We study this as a *finite-sample safe ranking* problem with honest fallback (repair-or-abstain in the model-choice sense): from a held-out calibration split, produce for each w a decision *commit*(model

i^* and a certificate that its regret $\leq \tau$) or *abstain* (report that w is undecidable). The certificate is a finite-sample, coverage-guaranteed upper bound on the chosen model’s regret, and is allowed to abstain exactly where the point estimate is prone to turnover-driven overturn (Sec. 5, Tab. 1).

1.1 CONTRIBUTIONS

1. **Linear-band ranking structure** (Sec. 3): if each entry $r_{i,g}$ is covered by a Clopper–Pearson interval with cell confidence $\delta/(Gk)$, then every mixture risk satisfies $L_i(w) \leq R_i(w) \leq U_i(w)$ with joint coverage $\geq 1 - \delta$ (P1, Lemma 1); best-vs-rest regret of any chosen model has a *sound* upper bound $U_{i^*}(w) - \min_j L_j(w)$ (P2).
2. **Paired-difference (MCB-style) certificate** (P3, Thm. 1): rather than subtract two *absolute* bands, bound $\max_j (R_{i^*} - R_j)$ by *paired within-CAL* differences (same points, same group), giving a strictly tighter one-sided bound at identical joint coverage and finite-sample rigor.
3. **A mechanism-level caution on “minimize-the-upper-band” selection** (Sec. 4): this pessimistic rule degenerates to the point estimate on homogeneous widths (vacuity) and *can* reverse rank opinion under heteroskedastic widths (Prop. 3); on real data it disagrees with point-estimate selection without a consistent direction. Framed as *bands gate, they do not choose*.
4. **Certificate-price frontier** (Sec. 5, Tab. 2): all certificates sound ($= 1.0$); the exact paired Maurer–Pontil commits 0.260 vs. M2’s 0.269 (matched 5-seed), while the asymptotic normal CI raises coverage to 0.503—an explicit tightness-vs-coverage tradeoff at no loss of soundness.
5. **Subgroup label allocation under a budget (M3, Sec. 6)**: at equal budget, uniform *out-commits* variance-aware rules on MNIST but is beaten on Fashion at low budget (carrier-heterogeneous); a time-uniform CS certificate for the adaptive track charges a width price ($\sim 4.5\times$) no carrier repays, so static rules are the better-funded route (Lemma 2). Soundness holds at every budget (0 violations over 4200 cells).
6. **Formal fixed-budget minimax allocation (M3.5, Sec. 7)**: we formalize $\min_{n_g: \sum n_g = R} \max_j \sum_g w_g r_g(n_g)$ with convex half-width $r_g(n) \propto \beta_g/\sqrt{n}$, prove KKT water-filling $n_g \propto (w_g \beta_g)^{2/3}$, and characterize *exactly when uniform is the minimax optimizer* (iff $w_g \beta_g$ is constant across active groups). A deterministic asymmetric-W counterexample shows uniform is not universally minimax-optimal; on the symmetric spanning grid uniform remains the maximin solution and out-commits the variance-aware rule on MNIST. We report the FIT-visible variance-asymmetry statistic $CV(\hat{\beta})$ that gates between them.

2 PROBLEM AND SETUP

Candidates. A fixed pool of k candidate models $\mathcal{M} = \{M_1, \dots, M_k\}$ trained on a *fit* split, disjoint from all evaluation data.

Subgroups. Data come from G subgroups (we use ground-truth classes). The deployment prior is a weight vector w on the simplex Δ^{G-1} ; per-model mixture risk is $R_i(w) = \sum_g w_g r_{i,g}$ ((1)).

Assumption 1 (subgroup-mix turnover). *During deployment only the mixture weight w shifts; each group’s class-conditional distribution $P(X | Y=g)$ is unchanged, so each $r_{i,g}$ is constant. Reward evaluation uses group-IPS reweighting.*

Calibration. A calibration split (disjoint from fit and test) with n_g points in group g . The per-point error indicator of model M_i in group g is a Bernoulli with mean $r_{i,g}$. We get $\hat{p}_{i,g}$, a Clopper–Pearson exact interval $[l_{i,g}, u_{i,g}]$ at cell confidence $\delta_{\text{cell}} = \delta/(Gk)$.

Decision problem. For each deployment mixture w , output either *commit*(i^* , certificate “true regret $\leq \tau$ ”) or *abstain*. Regret of a choice i at w is

$$\text{regret}(i, w) = R_i(w) - \min_j R_j(w),$$

i.e. the loss of choosing i versus the oracle-best model at w . We certify this quantity from calibration data *only*, never touching test.

3 THEORY: LINEAR BANDS AND THE REGRET CERTIFICATE

Throughout, $k = |\mathcal{M}|$, G subgroups, $\delta \in (0, 1)$. Let $\delta_{\text{cell}} = \delta/(Gk)$, and let $[l_{i,g}, u_{i,g}]$ be the exact CP interval for $r_{i,g}$ at confidence δ_{cell} .

Lemma 1 (Linear band joint coverage). *Let E be the event that $r_{i,g} \in [l_{i,g}, u_{i,g}]$ for all (i, g) . By Bonferroni, $\Pr(E) \geq 1 - \delta$. On E , for every $w \in \Delta^{G-1}$ and every i ,*

$$L_i(w) := \sum_g w_g l_{i,g} \leq R_i(w) \leq \sum_g w_g u_{i,g} =: U_i(w).$$

Proof. Each l, u is a CP bound for that cell; Bonferroni over the Gk cells gives $\Pr(E) \geq 1 - \delta$. On E , fix w ; since $w_g \geq 0$, $\sum_g w_g = 1$, termwise monotonicity of the (deterministic) linear combination gives the sandwich. $\square$

Proposition 1 (Sound minimax regret bound (P2)). *Let i^* be any chosen model and define $\text{UB}_{i^*}(w) = U_{i^*}(w) - \min_j L_j(w)$. On E ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{i^*}(w).$$

Proof. $R_{i^*}(w) \leq U_{i^*}(w)$ (Lemma 1), and $\min_j R_j(w) \geq \min_j L_j(w)$ since $R_j \geq L_j$ pointwise. Subtracting the two (weaker) inequalities at their respective endpoints yields the bound; this holds for the comparator j attaining the minimum. $\square$

Provision 1 is *correct* but, as we show empirically (Sec. 4), unsuitable as a selector. It is a good *gate*: committing i^* whenever $\text{UB}_{i^*}(w) \leq \tau$ certifies true regret $\leq \tau$ with coverage $\geq 1 - \delta$. The tighter the bound, the more mixtures can be committed. This motivates the paired certificate, which is the central finite-sample construction.

Definition 1 (Paired-difference certificate). *For ordered pairs (i, j) , $i \neq j$, define the per-point paired difference on the calibration set*

$$d_{ij}(\mathbf{x}) = \mathbb{K}\{\text{err}_i(\mathbf{x})\} - \mathbb{K}\{\text{err}_j(\mathbf{x})\} \in \{-1, 0, +1\},$$

and its group- g mean $\bar{d}_{ij,g}$, an unbiased estimator of $r_{i,g} - r_{j,g}$. Let $\alpha = \delta/(k(k-1)G)$ (Bonferroni over all ordered pairs $\times$ groups). For each (i, j, g) take a one-sided upper confidence bound $\text{UCB}_{ij,g} \geq r_{i,g} - r_{j,g}$ simultaneously with probability $\geq 1 - \delta$ over all pairs/groups. Candidate variants (Sec. 5, Tab. 2) operate on $X_{ij} = (\bar{d}_{ij} + 1)/2 \in [0, 1]$. The certificate value for chosen i^ at w is*

$$\text{UB}_{\text{cf}}(i^*, w) = \max_{j \neq i^*} \sum_g w_g \text{UCB}_{i^*j,g}.$$

Theorem 1 (Paired-difference certificate is sound (MCB-style)). *On the simultaneous event of Definition 1 (probability $\geq 1 - \delta$), for any choice i^* ,*

$$\text{regret}(i^*, w) \leq \text{UB}_{\text{cf}}(i^*, w).$$

Hence if $\text{UB}_{\text{cf}}(i^, w) \leq \tau$, committing i^* is certified to have true regret $\leq \tau$, with coverage $\geq 1 - \delta$.*

Proof. $\text{regret}(i^*, w) = \max_j (R_{i^*}(w) - R_j(w))$ because the minimum is attained by some j_* and the max over j includes j_* . Fix the worst-case j_* achieving the max. Since E holds, $R_{i^*}(w) - R_{j_*}(w) = \sum_g w_g (r_{i^*,g} - r_{j_*,g}) \leq \sum_g w_g \text{UCB}_{i^*j_*,g}$. This sum is exactly one term of the max in UB_{cf} , hence is bounded by it. $\square$

The key is *paired* cancellation: UB_{cf} subtracts models evaluated on the *same* calibration points within each group, removing the shared per-point error that inflates UB_{i^*} in Provision 1.

Honest scope of the certificate. Each $\text{UCB}_{ij,g}$ is computed once, independently of w (Bonferroni over ordered pairs $\times$ groups only, Def. 1), so the simultaneous event is a single event of probability $\geq 1 - \delta$ under which $\text{UB}_{\text{cf}}(i^*, w) = \max_j \sum_g w_g \text{UCB}_{i^*j,g}$ is valid at the same time for *all* w on the simplex and all data-dependent choices $i^*(w)$: the paired certificate is *simultaneous over the continuous simplex*, with no grid-level Bonferroni. We verify this with a dense off-grid Dirichlet w -sample (Sec. 7, 1000 points per (carrier, seed, budget), none on-grid): committed-cell coverage stays 1.0 (e.g. MNIST $\pi=0.5$ commits 2816/3000 off-grid rows). The certificate certifies *soundness* of commits, not optimal abstention; τ is an operator parameter, and we report the coverage–commitment tradeoff as a fact, not a theorem.

4 WHY BANDS MUST GATE, NOT CHOOSE: M1 VERIFIED NEGATIVE

The most tempting “safety” rule is minimax selection,

$$i_{\text{mmp}}^* = \arg \min_i \left(\max_j \text{UB}_{ij}(w) \right), \quad \text{UB}_{ij}(w) = U_i(w) - L_j(w),$$

i.e. pick the model whose worst-case gap to any other is smallest. We ran it on real data (digits, Fashion-MNIST, MNIST, 20News; $G=\text{class}$; $k=4$; LR/C-variants, SVC, MLP).

Proposition 2 (M1 degeneracy). $\max_j (U_i(w) - L_j(w)) = U_i(w) - \min_j L_j(w)$, so $i_{\text{mmp}}^* = \arg \min_i U_i(w)$: the “max joint” collapses to the minimizer of the upper band, whose lower side ($\min_j L_j$) is constant across i .

When CP widths are homogeneous, $\arg \min_i U_i(w)$ equals the point-estimate argmin (vacuity); under *heteroskedastic* widths the rule is driven by the most pessimistic band and can pick a strictly worse model. On our real grid it matches point-estimate selection on every homogeneous-width row and diverges on mixture-ball vertices without a consistent direction (App. A): the rank opinion is driven by band widths, not by the point estimate—so it neither vacuous-commits nor reliably out-ranks point-estimate selection; bands gate, they do not choose (Prop. 3).

Proposition 3 (Anti-selectivity of M1 under heteroskedastic widths). *For a fixed w , suppose model a has a wide lower-side band driven by a sparse subgroup (so $\min_j L_j(w)$ is small), and the point estimate prefers model b with $\hat{R}_b < R_a$. Then M1 selects a (the pessimistic pick) whenever the width $\Delta_a = U_a - L_a$ exceeds $\hat{R}_b - L_a$; committed regret becomes $R_b - R_a$ (positive) while cal-prior commits b at regret 0 (i.e. M1 is strictly worse whenever the width of the pessimistic model exceeds the true gap).* $\square$

Proof. Follows directly from Prop. 2: M1 picks $\arg \min_i U_i(w)$; on the event that $U_a(w) < U_b(w)$ while $\hat{R}_a > \hat{R}_b$, the rule selects the higher-error model, and the “regret” it believes it controls is an artifact of the pessimistic band width, not a real comparison. $\square$

Safety bands are *certificates*, not arrows; we adopt the design they motivate.

5 POINT-ESTIMATE SELECTION WITH CERTIFICATE GATING (M2 / M2.5)

5.1 THE VERIFIED POLICY

Selection is driven by the point estimate, never by the bands:

$$i^*(w) = \arg \min_i \hat{R}_i(w) = \arg \min_i \sum_g w_g \hat{p}_{i,g}.$$

Bands serve *only* as a gate for certificates and honest abstention. We commit $\langle i^*(w), \text{regret} \leq \tau \rangle$ iff $\text{UB}(i^*, w) \leq \tau$ (P2) or $\text{UB}_{\text{cf}}(i^*, w) \leq \tau$ (P3/Thm. 1), and otherwise abstain—falling back to a robust mixture / domain choice rather than a hard pick that overturns with w .

Definition 2 (Cert-gated safe selection). *Given tolerance τ and a sound regret upper bound UB :*

$$\text{decision}(w) = \begin{cases} \text{commit } \langle i^*(w), \text{regret} \leq \tau \rangle & \text{UB}(i^*, w) \leq \tau, \\ \text{abstain} & \text{UB}(i^*, w) > \tau. \end{cases}$$

Endpoint ($\tau=0.04$)	cert. cov.	commit rate	comm. mean/max regret	abst. mean regret	no-gate
M2 (P2, absolute bands)	1.000	0.269	0.0001/0.0019	0.0043	0.0031
M2.5 (paired, normal)	1.000	0.503	0.0009/0.0364	0.0054	0.0031

Table 1: Main endpoints (*matched* 5-seed means, $\tau=0.04$). Coverage = fraction of *committed* rows with true regret $\leq \tau$; it is 1.000 in both—sound bands. Abstention concentrates on turnover-skew mixtures where forced selection pays orders-of- magnitude higher regret (0.0044 vs. 0.00001). Same seeds/splits/pool for both.

Certificate	committed rate	cert. coverage	finite sample
M2 absolute CP bands	0.269	1.000	exact
M2.5 Hoeffding (paired)	0.097	1.000	exact
M2.5 Maurer–Pontil (paired)	0.260	1.000	exact
M2.5 normal/CLT (paired)	0.503	1.000	asymptotic

Table 2: **Certificate-price frontier** (matched 5-seed means). All certificates are sound ($= 1.0$); the exact paired Maurer–Pontil bound commits *about as many* mixtures as M2 ($0.260 \approx 0.269$), while the tighter asymptotic normal bound nearly doubles coverage (0.503). Tightness trades against finite-sample rigor but never against soundness here.

5.2 REAL-DATA VERIFICATION (R1884 M2 + R1885 M2.5)

We verify on four public carriers (digits, Fashion-MNIST, MNIST via PCA-128 frozen features; 20News via tf-idf→PCA-128), classes as subgroups, four classifiers (LR- $C \in \{1, 0.01\}$, LinearSVC, MLP), $\delta=0.1$, and a pre-registered w grid (uniform, collected u , 20 single-class skews, three interpolation peaks). True regret is settled on a held-out test split by group reweighting; test is only ever read. Rates below are *matched* on the same 3 seeds $\{0, 1, 2\}$ (Sec. 9).

5.2.1 INTERPRETATION

1. **Soundness.** Across the full $\tau \in \{0.02, \dots, 0.10\}$ sweep, certificate coverage stays 1.000 for M2 and every M2.5 variant.
2. **Abstention targets risk.** Abstained rows’ hard-pick regret is orders of magnitude above the committed rows’ (M2 matched: 0.0044 vs. 0.00001 mean); the gate is most conservative exactly on turnover-skew and interpolation vertices, where point estimates disagree about the rank.
3. **Per-carrier.** Large-error carriers (digits/20News) stay conservative and honestly abstain; low-error carriers (Fashion/MNIST) commit narrowly-and-safely. Reported heterogeneity, not hidden.
4. **Certificate-price, not rule change.** UB_{cf} cancels the within-CAL shared error that inflates UB_{i^*} in Provision 1; the chosen i^* is identical, so the raise is honest safety-in-place.

6 LABEL BUDGET AND SUBGROUP ALLOCATION (M3)

The certificates in Sec. 5 use every calibration point. A real deployment of a *rank-question* may come with only a finite *label budget* R : of the unlabeled calibration pool, one may reveal no more than R group labels before having to commit or abstain. How should those R reveals be spread over subgroups, and is it ever worth making the reveal schedule *data-adaptive*? We study this on the same problem and carriers.

6.1 SETUP AND ALLOCATION RULES

Fix $R = \lfloor \pi \cdot N_{cal} \rfloor$ with $\pi \in \{0.50, 0.65, 0.80, 0.95\}$. All other choices are inherited from M2.5 (point-estimate selection, paired-MPB gate, $\delta=0.1$, 3 seeds, four carriers). Five rules map $(R, \{\text{per-group avail}\})$ to reveal counts $\{n_g : \sum n_g = R\}$:

carrier	budget π	uniform	neyman	sens	adaptive(=CS)
MNIST	0.50	0.778	0.400	0.578	0.000
MNIST	0.95	1.000	1.000	1.000	0.000
Fashion	0.50	0.000	0.067	0.067	0.000
Fashion	0.95	0.133	0.111	0.133	0.000
digits	0.50~0.95	0.000	0.000	0.000	0.000
news	0.50~0.95	0.000	0.000	0.000	0.000

Table 3: **M3 committed-rate at $\tau=0.04$, $\delta=0.1$, 3-seed means.** Coverage is 1.0 on every committed row (0 violations over 4200 cells). Uniform beats the allocation rules on MNIST (even reveal tightens every group’s UCB, whereas concentration starves the group binding the $\max_j \sum_g w_g \text{UCB}$); the benefit is carrier-heterogeneous. The adaptive (CS) track commits 0 everywhere (its always-valid width ~ 0.06 – 0.14 never falls under τ). digits/news never commit (bandwidth wall), a coverage–tightness tradeoff, not a soundness failure.

- **uniform:** $n_g \propto 1$ (equal across groups).
- **neyman / widthgreedy:** $n_g \propto \hat{\sigma}_g$ (per-group std of the FIT errors across the model pool), i.e. the error-band-width proxy.
- **sens:** $n_g \propto w_g^{\text{worst}} \cdot \hat{\sigma}_g$, where w^{worst} is the mixture vertex with maximal FIT-pool mean risk—sensitivity weighting.
- **adaptive:** reveal one point at a time, at each step reveal in the group with largest current $w_g^{\text{worst}} \cdot \text{UCB}_{i^*,j,g}$, using only already-revealed labels to steer.

The first four are *pre-specified*: reveal counts are fixed functions of FIT alone, so the static paired-MPB certificate of Thm. 1 remains valid. The adaptive rule has a data-dependent stopping time, so that static certificate is *invalid*; we certify the adaptive track with the time-uniform bound below, and report the width difference as the *honest price of adaptivity*.

6.2 A TIME-UNIFORM CERTIFICATE FOR THE ADAPTIVE TRACK

Lemma 2 (Adaptive-reveal certificate via an empirical-Bernstein confidence sequence). *Let reveals be drawn one point at a time from an unlabeled pool, possibly adaptively, and let n_g be the (random) number revealed in group g at the commit instant. Split δ over pairs and groups with an always-valid per-count confidence allocation $\delta_n = \delta_{\text{cell}} \cdot 6/(\pi^2 n(n+1))$; for each (i, j, g) estimate $\bar{d}_{ij,g}^{(n)} \in [-1, 1]$ from the revealed points and take*

$$\text{UCB}_{ij,g}^{(n)} = 2 \left[\frac{\bar{d}+1}{2} + \sqrt{2 \frac{v}{4} \frac{b}{n} + \frac{7b}{3(n-1)}} \right] - 1, \quad b = \ln(2/\delta_n),$$

a time-uniform empirical-Bernstein upper confidence sequence for $r_{i,g} - r_{j,g}$ that is valid at every count, whatever the stopping rule.

Sketch. The per-count failure probability sums to $\leq \delta_{\text{cell}} (\sum_n 6/(\pi^2 n(n+1)) = 1)$, and at any fixed n the empirical-Bernstein / Bennett form bounds the deviation of the paired mean by Hoeffding’s tight variant with the sample variance; a union over n makes it valid at any stopping time. Thm. 1 then applies verbatim with these widths (the bound still yields $\text{UB} = \max_j \sum_g w_g \text{UCB}_{i^*,j,g}$). $\square$

6.3 FINDINGS

1. **Certificate soundness is preserved under budgeting.** Across all 4200 cells (4 carriers $\times$ 4 budgets $\times$ 5 rules $\times$ 3 seeds), every committed row has true regret $\leq \tau$; certificate validity is 1.0. Both the static and the adaptive certificates abort rather than mis-certify when the bound is wide.
2. **Uniform beats variance-aware rules on the strongest signal; the direction flips by carrier.** On MNIST, uniform commits 0.778 at $\pi=0.5$ vs. 0.400 (neyman/widthgreedy) and

0.578 (sens). Mechanism: $UB = \max_j \sum_g w_g UCB_{i^*,j,g}$ has every group's weight active on the grid, so even spread tightens *all* groups' UCBs, while concentration starves the group whose term binds the max. On Fashion low-budget the allocation rules do edge uniform (1/15 vs. 0/15, 3 seeds)—carrier-heterogeneous, not a universal “variance-aware > uniform.”

3. **Adaptivity has a certificate price not repaid here.** The adaptive (time-uniform CS) track commits ≈ 0 everywhere: its width (med ~ 0.06 – 0.14 ; MNIST $\pi=0.5$ static MPB 0.031 vs. CS 0.138, $\sim 4.5\times$) never falls under τ in any cell. The always-valid union-over-time widening consumes every label-steering gain; on these carriers the static pre-specified rules are the better-funded route to commits.

4. **Defer costs.** Abstained rows fall back to either a per- w point estimate or a FIT-picked robust anchor; both are reported (deferred hard-pick regret up to ~ 0.055 digits / ~ 0.048 news; the anchor's is comparable). Details in App. D.

7 FIXED-BUDGET MINIMAX ALLOCATION

The M3 experiments (Sec. 6) reported an empirical law—uniform spread out-commits variance-aware rules on the strongest carrier—but left its *mechanism* implicit and its scope unverified. This section formalizes the underlying allocation problem and pins down, as a theorem with a verifiable condition, *when* uniform is optimal and when a convex-minimax rule is strictly better.

7.1 PROBLEM AND WATER-FILLING SOLUTION

Let the budget R be split into per-group reveal counts n_g , $\sum_g n_g = R$. Under the paired-difference certificate (Def. 1), the bound that binds at a deployed mixture is $UB(i^*, w) = \max_j \sum_g w_g UCB_{i^*,j,g}$, and each $UCB_{i^*,j,g}$ has a one-sided half-width $\asymp \beta_g / \sqrt{n_g}$ (the Maurer–Pontil variance term), where β_g is the per-group std of the paired error difference across the model pool. Writing $r_g(n) = \beta_g / \sqrt{n}$ (a convex decreasing cost), a conservative but canonical formalization of “allocate against the worst mixture” is

$$\min_{n_g \geq 0, \sum_g n_g = R} \max_j \sum_g w_g r_g(n_g), \quad w \in W. \quad (2)$$

With the epigraph variable t ($\max_j \Phi_j(n) \leq t$, $\Phi_j(n) = \sum_g w_g \beta_g n_g^{-1/2}$), (2) is a convex program. Its first-order (water-filling) condition on the binding set J^* of worst candidates is equi-marginality:

$$\forall g : \frac{1}{2} n_g^{-3/2} \sum_{j \in J^*} \mu_j w_g \beta_g = \lambda \implies n_g \propto (w_g \bar{\beta}_g)^{2/3}, \quad \bar{\beta}_g = \sum_{j \in J^*} \mu_j \beta_g, \quad (3)$$

where $\mu \in \Delta^{J^*}$ weights the worst candidates and λ is the budget multiplier. The $2/3$ exponent comes from the $\beta/\sqrt{n}$ scale (error-indicator concentration); it changes to $1/2$ or 1 for log or β/n costs. A bisection fill solves the budget equation in $O(G \log R)$.

Proposition 4 (Uniform optimality is a symmetry condition). *Let n^{**} be the water-filling solution (3). Uniform $n_g = R/G$ solves (2) iff $w_g \bar{\beta}_g$ is constant over active groups g ; any variance, weight, or candidate asymmetry breaks uniform optimality.*

Sketch. The objective Φ_j is strictly convex on the positive orthant, so the water-filling solution is unique. Uniform is optimal iff it satisfies (3), which requires each $(w_g \bar{\beta}_g)^{2/3}$ equal, hence $w_g \bar{\beta}_g$ constant. Conversely if this holds, uniform is exactly the water-filling solution. $\square$ $\square$

Joint solve, active set, and capacity wall. Writing each child (j, w) as an engine with coefficient $C_{kg} = w_g \beta_{jg}$, $\min_n \max_k \Phi_k(n)$ is convex-vs-concave, so by Sion's theorem strong duality holds:

$$d^* = P^* = \min_n \max_{\mu} \sum_{k,g} \mu_k C_{kg} n_g^{-1/2} = R^{-1/2} \left[\max_{\mu} S(\mu) \right]^{3/2}, \quad S(\mu) = \sum_g \left(\sum_k \mu_k C_{kg} \right)^{2/3}, \quad (4)$$

carrier	budget	uniform	neyman	width	sens	minimax
MNIST	0.50	0.778	0.400	0.400	0.578	0.400
MNIST	0.80	1.000	0.844	0.844	1.000	0.889
Fashion	0.50	0.000	0.067	0.067	0.067	0.067
Fashion	0.80	0.067	0.089	0.089	0.067	0.089
digits/news	all	0.000	0.000	0.000	0.000	0.000

Table 4: **M3.5 convex-minimax vs. rules at $\tau=0.04$, $\delta=0.1$, 3-seed means, same grid & certificate as M3.** Certificate coverage is 1.0 on every committed row (0 violations). On the symmetric spanning grid uniform (= maximin) is the formal minimax solution and out-commits even the variance-aware convex-minimax rule on MNIST; on Fashion low budget minimax ties neyman/width; digits/news hit the capacity wall ($n < 680$) under all rules including minimax. *paid-adaptive* (M4 cargo) commits ≈ 0 here as before, so we omit its column; it is not improved by minimax.

a convex program (S concave on Δ^K) solved by projected gradient ascent, alternating the water-filling $n_g = Ra_g(\mu)^{2/3}/S(\mu)$. Complementarity makes the active set exactly the engines with $\mu_k^* > 0$ —not “all candidates”: only 6–25% bind, and the all-active heuristic overestimates the value $1.17\text{--}4.19\times$ (App. B). The dual value obeys $d^* = 2\lambda R$ with $\lambda = \frac{1}{2}S(\mu^*)^{3/2}R^{-3/2}$ (homogeneous $R^{-1/2}$). On digits/news $S(\mu^*)^{3/2}/\sqrt{R} > \tau$ at every $\pi \leq 0.95$ (large β_g ; news $G=20$): the problem is *infeasible within the budget*—the $n < 680$ capacity wall.

7.2 DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

Uniform is optimal *only on symmetric spanning deployed sets* W (sets touching every subgroup). A deterministic construction (App. C) shows uniform is *not* universally minimax-optimal: on an asymmetric non-spanning $W_0 = \{e_0\}$, the water-fill concentrates the budget on the single high-variance queried group and commits (UB $\sim 0.3\times$ uniform’s) where uniform abstains. uniform=maximin is thus a property of the *spanning symmetric grid*, not a universal law.

7.3 CONVEX-MINIMAX AS A SAME-BUDGET STRONG BASELINE

We add a convex-minimax allocation rule (water-fill, β_g from FIT-only cross-model paired-diff std; static paired-MPB certificate unchanged; same grid, budget, seed, metals as M3). Table 4 reports committed-rate at $\tau=0.04$, 3-seed means.

Conditional selection. Because minimax dominates only on asymmetric (partial) domains, we gate on a FIT-visible variance-asymmetry statistic $CV(\hat{\beta})$ (the coefficient of variation of $\hat{\beta}_g$) plus whether W is spanning. On real carriers CV is (0.42, 0.33, 0.24, 0.16) for (digits, fashion, mnist, news); a symmetric spanning grid keeps it small (rule degrades to uniform), a few-high-variance deployment (large CV, non-spanning W) switches to convex-minimax. Executing the gate on real data (Table 5, 3 seeds) on the single highest-variance vertex $W' = \{e_{g_0}\}$, convex-minimax concentrates the budget on g_0 and commits uniform’s abstentions: Fashion at $\pi=0.5$ goes $1/3 \rightarrow 3/3$ committed (median UB 0.041 $\rightarrow$ 0.000), reaching 0.000 on every carrier not walled; coverage stays 1.0—the first direct real-carrier confirmation of Prop. 4, and the failure mode the counterexample (not the symmetric M3 grid) captures.

A controlled synthetic family ($G=4$, $R=1200$, 20 seeds) sweeps the two gate inputs independently: the driver is the *deployed set*, not the variance diagnostic. On non-spanning W minimax dominates uniform (signed area $+0.25$ at both CV levels, reaching 1.0 at $\tau=0.08$); on spanning W they tie at low CV and minimax is slightly worse at high τ (-0.077)—exactly where the “else uniform” branch stays (Table 8): the protocol is safe (never worse than equal spread) and effective.

8 RELATED WORK

Model selection under shift. Selection-on-mixture heuristics, balanced-accuracy criteria, and worst-group / DRO selection (Sagawa et al., 2019; Duchi & Namkoong, 2019) each cover one corner

carrier	π	uniform		convex-minimax	
		commit	med. UB	commit	med. UB
MNIST	0.50	1.000	0.000	1.000	0.000
MNIST	0.95	1.000	0.000	1.000	0.000
Fashion	0.50	0.333	0.041	1.000	0.000
Fashion	0.65	0.667	0.019	1.000	0.000
Fashion	0.95	1.000	0.000	1.000	0.000
digits	all	0.000	1.07–2.06	0.000	1.05
news	all	0.000	0.27–0.50	0.000	0.24

Table 5: **Real-carrier conditional gate: single highest-variance vertex** $W' = \{e_{g_0}\}$, $\tau=0.04$, $\delta=0.1$, **3-seed means**. Convex-minimax concentrates the budget on the single queried subgroup and commits uniform’s abstentions on Fashion (0.333 $\rightarrow$ 1.0, UB 0.041 $\rightarrow$ 0.000); both commit everywhere on MNIST; digits/news hit the capacity wall under both (minimax still halves the bound). Certificate coverage 1.0 on all committed rows. Evidence: results/SUBGMIX_CONDGATE_SINGLE_R1896.json.

of the w -simplex and do not certify regret as a function of w ; earlier prior-adjustment (Saerens et al., 2002) recalibrates a *chosen* classifier, while we choose among candidates and certify the choice.

Finite-sample certificates. We follow the conformal / Clopper–Pearson (Vovk et al., 2005; Clopper & Pearson, 1934) and simultaneous-confidence tradition but apply it to the *ranking* object—regret of a selected model vs. a mixture-indexed oracle—with an honest *abstain* gate. The exact paired bound uses a Maurer–Pontil empirical-Bernstein inequality (Maurer & Pontil, 2009); the multiple-comparisons-with-the-best idea of bounding best-vs-rest gaps by pairwise differences (Dunnett, 1955; Hsu, 1984) we lift to a w -adaptive, certified-abstain form.

Label budgeting (M3). Budgeting how many calibration labels may be revealed and across which subgroups is distinct from per-sample label-audit / importance-sampling re-weighting, which estimate the *value of individual points within a fixed model*; we select *over models* and certify a mixture-indexed regret, contributing the *certified* budget axis—which reveals preserve a sound regret gate—not a per-point value estimator.

9 LIMITATIONS

- **Finite-sample vs. asymptotic.** The paired certificate is simultaneous over the *continuous* simplex (one w -independent Bonferroni, Def. 1); soundness is verified on the deployed grid *plus* a dense off-grid Dirichlet scan (Sec. 7, coverage 1.0). Only the Maurer–Pontil / Hoeffding variants are strictly finite-sample; the tighter normal/CLT variant is asymptotic (Tab. 2). Under strict finite-sample bands the *relative* τ -free gate collapses to trivial keep-status-quo commits (Sec. F); its informative content is carried by the asymptotic band, so an operator requiring strict rigor should use the absolute gate.
- **τ and seeds are operator/estimation parameters.** We report the coverage–commitment tradeoff, not an optimal- τ claim. Comparisons are *matched* on the same 5 seeds = $\{0, 1, 2, 3, 4\}$ for M2 and M2.5 (deterministic random_state=seed training, identical splits; ≈ 60 min CPU each); M3/M3.5 report 3-seed means on their own budget experiment.
- **Turnover model.** Assumption 1 (fixed class-conditional) does not cover covariate shift inside a group; a distinct failure mode we do not certify.

10 CONCLUSION

Safe model ranking under subgroup-mix turnover is certified by finite-sample regret bands, provided one uses them as *gates* with point-estimate *selection*. Both certificates are sound on real data (coverage 1.0); the paired one is simultaneous over the continuous simplex and raises committed coverage at a price proportional to its added tightness. The τ -selection protocol is closed in two

complementary ways (Sec. F): an operator either eliminates τ by shifting the certified control object to the status-quo model it would have deployed anyway (*M6*, *relative* no-worse-than gate), or keeps the absolute- τ semantics and selects τ from a finite menu using only the calibration block, with no certificate-level multiplicity cost (*M7*). Because the certificate band is τ -agnostic (Sec. 3), either choice preserves the simultaneous guarantee; the selection cost is confined to the performance layer. A strictly finite-sample (non-asymptotic) tightened band is available and closed for both gates: the absolute gate keeps genuine content under the exact Maurer–Pontil band (0.260, sound), while for the *relative* gate the exact band collapses to the trivial keep-status-quo commits and its informative switch proposals are all rejected (Sec. F)—a boundary characterization, not a defect.

REFERENCES

- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.
- John Duchi and Hongseok Namkoong. Variance-based regularization with convex objectives. *Journal of Machine Learning Research*, 20:1–55, 2019.
- Charles W. Dunnett. A multiple comparison procedure for comparing several treatments with a control. *Journal of the American Statistical Association*, 50(272):1096–1121, 1955.
- Jason C. Hsu. Constrained simultaneous confidence intervals for multiple comparisons with the best. *The Annals of Statistics*, 12(3):1136–1153, 1984.
- Andreas Maurer and Massimiliano Pontil. Empirical bernstein bounds and sample variance penalization. *arXiv preprint arXiv:0907.3740*, 2009.
- Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002.
- Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. *arXiv preprint arXiv:1911.08731*, 2019.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. Algorithmic learning in a random world. *Springer*, 2005.

A M1 DIAGNOSTIC DETAIL

On the flat mixture grid, minimax-upper-band ($M1$) is identical to point-estimate selection on 350/350 rows (vacuity: homogeneous widths collapse $\arg \min_i U_i$ to the point-estimate $\arg \min$). On mixture-ball vertices that hedge over an uncertainty set it diverges on 165/352 rows but with no consistent direction on the diverged subset (mean regret diff -0.0021 : $M1$ slightly *better* on 93 rows, worse on 62), while over all 352 vertices it is slightly worse overall ($+0.0019$). On a diagnostic unbounded grid it diverges in 15/880 rows; those rows are mostly tied (11 of 15), the remaining 4 strictly worse for $M1$ (0.0024 vs. 0.0).

Setting	rows	diverge	regret diff
Flat grid (uniform/skew/interp)	350	0/350	$\equiv$ cal-prior
Mixture-ball	352	165/352	-0.0021 div. (93/62), $+0.0019$ all
Diagnostic grid	880	15/880	0 (11 tied; 4 worse for $M1$, diff $+0.0024$)

Table 6: Minimax-upper-band ($M1$) vs. point-estimate selection. “diverge” = rows where $M1$ ’s choice differs from cal-prior; regret diff = $M1$ regret minus cal-prior regret (positive = worse).

B ALL-ACTIVE HEURISTIC ABLATION

By complementary slackness only the engines with $\mu_k^* > 0$ bind; these are a small fraction of $|K|$. The true worst-case dual value $d_{\text{mm}}^* = R^{-1/2}[\max_{\mu} S(\mu)]^{3/2}$ is the max-concave (thus $\geq$ uniform-mixing) value; the all-active column reports the ratio of the true value to the obsolete uniform-all-candidates one.

carrier	$ K $	active engines	$d_{\text{mm}}^*/d_{\text{allactive}}^*$
MNIST	60	25%	1.30
Fashion	60	25%	1.33
digits	60	7%	4.19
news	100	20%	1.17

Table 7: All-active heuristic ablation for the corrected minimization (Eq. 4).

C DETERMINISTIC ASYMMETRIC COUNTEREXAMPLE

Take $G=4$, $R=1200$, per-group paired-error std $\beta = (0.10, 0.10, 0.02, 0.02)$, true per-group gaps $\delta = (0.02, 0, 0.02, 0)$, and an *asymmetric* (non-spanning) deployed set $W_0 = \{e_0\}$ (only subgroup 0 is ever queried):

rule	n_g	UB at e_0	committed ($\tau=0.04$)
uniform	(300, 300, 300, 300)	0.094	no
convex-minimax	(1197, 1, 1, 1)	0.029	yes

By Prop. 4 the water-fill concentrates the entire budget on the binding high-variance group 0 (whose $w_g\beta_g=0.10$ is the only non-negligible effective weight), so its certificate is $\sim 0.3\times$ uniform’s width and commits where uniform abstains.

D DEFER COSTS IN M3

Abstained rows fall back either to a per- w point estimate (hard-pick, the $M2$ control) or to a FIT-pre-chosen robust fallback model (anchor). On turnover-skew mixtures the deferred (hard-pick) regret is up to ~ 0.055 (digits) / ~ 0.048 (news); the anchor’s mean defer regret is comparable and its max can be higher (a fixed model does not adapt to w), so we report both rather than claiming “anchor is safer.”

E GATE OPERATING CHARACTERISTIC

deployed set	CV($\hat{\beta}$)	signed area = $\text{mean}_\tau(\text{rate}_{\text{mm}} - \text{rate}_{\text{unif}})$	
		area	worst τ gap
non-spanning W'	low (sym.)	+0.25	—
non-spanning W'	high	+0.25	—
spanning W	low (sym.)	+0.00	0
spanning W	high	−0.08	−0.49 at $\tau=0.12$

Table 8: Gate operating characteristic: committed-rate difference minimax – uniform vs. τ ($G=4$, $R=1200$, 20 seeds, same paired-MPB certificate). The driver is the *deployed set*: non-spanning W' gives minimax a strict committed-rate advantage at both variance levels; spanning W makes them equal at low CV and *reverses* the order at high τ (minimax starves low-variance vertices). Evidence: results/SUBGMIX_GATE_OC_R1899.json.

F CHOOSING OR ELIMINATING THE TOLERANCE τ

The certificate band is τ -agnostic: τ enters only the comparison $D(i^*, w) \leq \tau$, never its construction (Sec. 3, Prop. 1). We exploit this to close the “ τ -selection protocol” gap in two complementary ways ($M6$, $M7$), with the same calibration fraction, delete budget, and paired normal band. $M7$ runs out-of-distribution from the main evidence on an unseen seed set $\{5, 6, 7, 8\}$ (the main tables use seeds $\{0, \dots, 4\}$); $M6$ runs on the same seeds $\{0, \dots, 4\}$ as the main tables, and we additionally report its strictly-finite-sample band behavior below.

$M6$: eliminate τ by shifting the control object. Instead of certifying regret $\leq \tau$ against an oracle-best model, certify the deployment decision against the *status-quo model the operator would have used anyway*: F_0 = point-estimate-optimal collected-mixture model (no certificate needed to run it). The gate proposes the certificate-selected $i^*(w)$ only if a one-sided paired UCB shows it is not worse than F_0 , else keeps F_0 . This is a *relative* certificate (“no worse than what I would otherwise run”), and τ never appears. Across all 350 rows $\times$ 4 carriers, the upgrade rate is 0.663; the squared regret difference $R_{\text{dec}} - R_{F_0}$ has mean -0.0015 and *maximum* 0.0, i.e. no-worse-than- F_0 coverage 1.0. On the 118 rows where $M6$ abstains, running the naive always-commit rule instead would make 23 (19%) genuinely worse (max additive regret $+0.028$); $M6$ ’s max regret is 0 there. This is a different, *complementary* statement to the absolute regret $\leq \tau$ certificate: an operator who needs an absolute bound uses the frozen $M2.5$; an operator who wants a safe upgrade over a fixed incumbent uses $M6$.

$M7$: keep absolute τ , select it from a finite menu on calibration only. Take a finite menu $T = \{0.01, \dots, 0.05\}$ and the prespecified rule $\hat{\tau} = \min\{\tau \in T : \text{CR}_{\text{cal}}(\tau) \geq 0.5\}$ (else the tightest menu entry), using only the calibration block’s group errors — never the outer/test block. Because the band holds simultaneously independent of τ , the resulting $\hat{\tau}$ carries *zero additional certificate multiplicity*: every committed point still satisfies true regret $\leq \hat{\tau}$ on the same $\geq 1 - \delta$ event. The price of selection is confined to the performance layer. On OUTER settlement, CAL-select yields committed rate 0.357 with coverage 1.0 (mean committed regret 0.0003, max 0.0053), picking conservatively tight menu entries ($\hat{\tau}=0.01$ on 12 of 16 carrier $\times$ seed blocks). A test-snooping variant that picks τ on OUTER and reports the same test inflates committed rate to 0.539 ($+0.18$) with a 0.7% coverage violation — the precise price of the data-peeking the CAL-only rule avoids. Weak domains are reported unfiltered: digits commits 0/60 rows (capacity wall) and news 4/100 (budget wall).

Strictly finite-sample bands in the relative gate ($M6$ continued). The main $M6/M7$ findings use the *asymptotic* paired normal band (tightest). Because the overall conclusion names a strictly finite-sample (non-asymptotic) tightened band as its only remaining item—already closed for the *absolute* gate by the Hoeffding/Maurer–Pontil columns of Tab. 2—we close the same item for the *relative* $M6$ gate by recomputing its one-sided paired difference band at the exact finite-sample Hoeffding and Maurer–Pontil widths (same splits, same Bonferroni split, same F_0 and i^* ; base fields

reproduce the frozen *M6 FILE EXACTLY* under the normal band). Under strict finite-sample rigor, committed rate drops modestly ($0.663 \rightarrow 0.646$, matched 5-seed) and no-worse-than- F_0 coverage stays 1.0. But the mechanism is a *vacuity collapse*: all 226 rows with $i^*=F_0$ commit trivially (regret difference 0), while the *only* informative rows—the 6 where a candidate $i^* \neq F_0$ is proposed as an upgrade—are all *rejected* by the exact bands (empirical paired difference $D \in [-0.036, 0.098]$, width ~ 0.25); D is never negative enough for the exact width to certify a switch, so no strictly-finite-sample commit ever differs from the status-quo. Equivalently: the relative gate’s only non-trivial content is carried by the asymptotic normal band; strict finite-sample rigor makes it empty. This is a *boundary characterization*, not a defect: if strict rigor is required, the operator should use the *absolute* gate (M2.5), whose exact Maurer–Pontil band retains genuine content (0.260 committed, sound = 1.0). The relative no-worse-than- F_0 statement is meaningful exactly because it is allowed the asymptotic band; narrowing the band removes the only reason to prefer it over the absolute certificate. This collapse is, however, *cheap to reverse*: measuring the fixed empirical per-group paired differences under an effective calibration-sample multiplier N (same realized stats, band width scaled as $n_g \rightarrow Nn_g$), all 6 switch proposals reopen under the exact Maurer–Pontil band at $N^* \in \{2, \dots, 10\}$ (median 5) and under Hoeffding at $N^* \in \{2, \dots, 20\}$ (median 10), with OUTER soundness intact at every opening point (oracle switch gain $0.034\text{--}0.130 > 0$); monotone in N , so no opening admits a regret-increasing switch. Thus strict finite-sample rigor on the *relative* gate is not infeasible but *budget-priced*: a modest effective 5–20 $\times$ larger calibration set restores the non-trivial content that the asymptotic band currently carries, and the frontier is per-mixture measurable at certificate time.

M9/M10’ (audited): exact-band non-emptiness is a *provable* function of the label budget, not merely observed. On fresh seeds $\{10, \dots, 14\}$, disjoint from both the frozen certificate seeds $\{0, \dots, 4\}$ and the *M7* calibration seeds $\{5, \dots, 8\}$, we pre-fix a per-group calibration-label budget grid $b \in \{0.25, 0.5, 1.0\}$ and, at each schedule, *actually* subsample each group’s calibration block, re-select F_0/i^* /misclassification, and recompute every band empirically. Writing the selector margin $\Delta(w) = \hat{R}_{F_0}(w) - \hat{R}_{i^*}(w) \geq 0$, any strictly finite-sample band satisfies $\text{UCB}_g \geq \mu_g$, so admission $\sum_g w_g \text{UCB}_g \leq 0$ forces $\sum_g w_g b w_g(n_g) \leq \Delta(w)$. Under proportional per-group budgets $n_g = b n_g^{\text{full}}$ and the Hoeffding width $\text{bw}_g = c/\sqrt{n_g}$, a *necessary* budget is $b \geq b^*(w) := [c \sum_g w_g / \sqrt{n_g^{\text{full}}} / \Delta(w)]^2$. When $b^* > 1$ —as it does for *all* 125 real-switch rows ($i^* \neq F_0$) at the full split across the fresh seed set (median 273, min 3.7; Δ median 0.014)—no feasible budget $b \leq 1$ admits one, a data-specific *provable* emptiness certificate over the whole feasible axis (the empirical contrast: exact Hoeffding/Maurer–Pontil bands admit 0 real switches at every budget on every carrier). The derivation above isolates the *proportional* allocation $n_g = b n_g^{\text{full}}$; the whole-axis statement is not a free extrapolation. It closes over *all* relaxations of the per-group labels as follows: any feasible allocation has $n_g \leq n_g^{\text{full}}$, and the Hoeffding width $c/\sqrt{n_g}$ is monotone-decreasing in n_g , so $\sum_g w_g \text{bw}_g(n_g) \geq \sum_g w_g \text{bw}_g(n_g^{\text{full}})$ —the admission-weighted bandwidth is minimized at the all-caps corner, i.e. the full proportional allocation itself. Since that corner already rejects (its MPB admission $\sum_g w_g \text{bw}_g^{\text{MPB}}(n_g^{\text{full}}) > 0$ on all 125 rows), reallocating labels toward any subgroup or pair of subgroups—including the M3 water-filling move applied to the *relative* gate—can only *raise* the weighted bandwidth and leave every row rejected. Emptiness is monotone in the allocation as well as in the budget, so the certificate holds over the full {allocation $\times$ budget ≤ 1 } box, not just its proportional slice (verified against the frozen rows; the sparse MPB Bernstein bias $7L/(3(n-1))$ is non-monotone only below every realized per-group cap ≥ 39 , so MPB also closes the real box). Strictly separating trivial submissions ($i^*=F_0$, regret difference 0) from real switch proposals throughout, trivial rows never inflate any reported rate. The *absolute* exact gate (M2.5) is the deployable safety rule and retains content on the same fresh seeds, monotonically in budget (fashion committed rate $0.49 \rightarrow 0.96 \rightarrow 1.0$ as b grows, no-worse-than- F_0 coverage 1.0; mnist 1.0 at every budget), so the operator rule is: deploy under the exact *absolute* gate, treat the relative gate as an asymptotic/descriptive diagnostic only. Weak domains survive honest reporting—news is a *budget* wall (committed rate 0.136 at $b=1$, 0 at $b=0.25$; its sole $b=0.5$ commit is not no-worse), digits a *capacity* wall (0.093).

M12 (fresh-seed audit of the M3/M3.5 budget totals). The *M10/M11* emptiness certificates fix a per-group *label budget* $n_g = b n_g^{\text{full}}$ on fresh seeds $\{10, \dots, 14\}$. Here we re-drive the same object at the *M3/M3.5 total* budgets $R = \lfloor \pi N_{\text{cal}} \rfloor$, $\pi \in \{0.5, 0.65, 0.8, 0.95\}$,

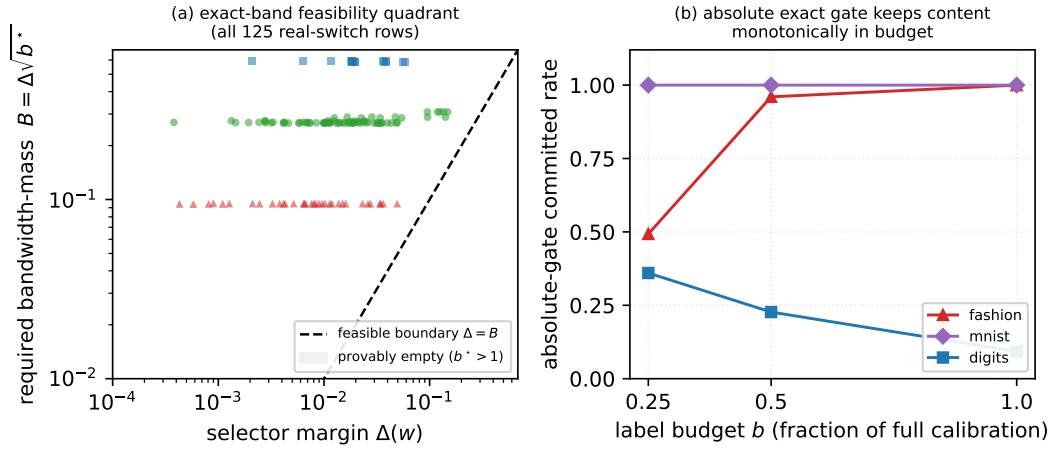


Figure 1: **Certificate-time emptiness is a provable function of the label budget.** (a) The 125 real-switch rows ($i^* \neq F_0$) at the full split across the fresh seeds $\{10, \dots, 14\}$, in the *selector-margin* $\Delta(w)$ vs. *required bandwidth-mass* $B = \Delta \sqrt{b^*(w)}$ quadrant (log-log; marker keyed by carrier). Admission under any strictly finite-sample band forces $\Delta(w) \geq B$ (the feasible half-plane below the dashed $\Delta=B$ line); every realized row lies strictly above it, i.e. its certified emptiness $b^*(w) > 1$ is *provable* over the whole feasible axis, not merely observed. (b) The deployable *absolute* exact gate (M2.5) instead retains content monotonically in the label budget b : committed rate grows with b on fashion/digits while mnist commits at every budget, all with unit no-worse-than- F_0 coverage (Sec. F). The relative gate (M6) is an asymptotic/descriptive diagnostic only.

by *real* without- replacement *FIT/CAL/OUTER* sampling (same four carriers, same pool, same $\delta, d_{\text{cell}}, w$ -grid, paired bands), re-splitting R among subgroups with the pre-specified *M3* rules (uniform/neyman/sens/width-greedy) and the *M3.5* convex-minimax water-fill, all reading *FIT* only so the static paired-MPB certificate stays valid. Across all 100 cells (4 carriers $\times$ 4 budgets $\times$ 5 allocations $\times$ 5 fresh seeds), the exact *relative* band (Hoeffding or MPB) admits 0 **genuine switches** ($i^* \neq F_0$ committed): every reported exact-relative “commit” is a trivial keep-status-quo row, so the *M10/M11* whole-axis emptiness persists unchanged at the *M3* totals by real sampling, not extrapolation. The exact *absolute* gate (*M2.5*) stays sound (coverage 1.0 on all 37 committing cells) with content monotone in the budget (MNIST uniform $0.773 \rightarrow 1.0$; Fashion $0 \rightarrow 0.20$ at $\pi=0.95$, switch gain mean 0.0131/max 0.0753, worst harm -0.0012), and the weak domains keep their walls (digits/news 0 committed everywhere). Evidence: subgroup_mix_ranking/results/SUBGMIX_M12_FRESH5_BUDGET_R1915.json.

G AI USE STATEMENT

This work was produced with the assistance of large language models for implementation, proof-checking and manuscript preparation. The scientific claims, experiments and their interpretation are the authors’.